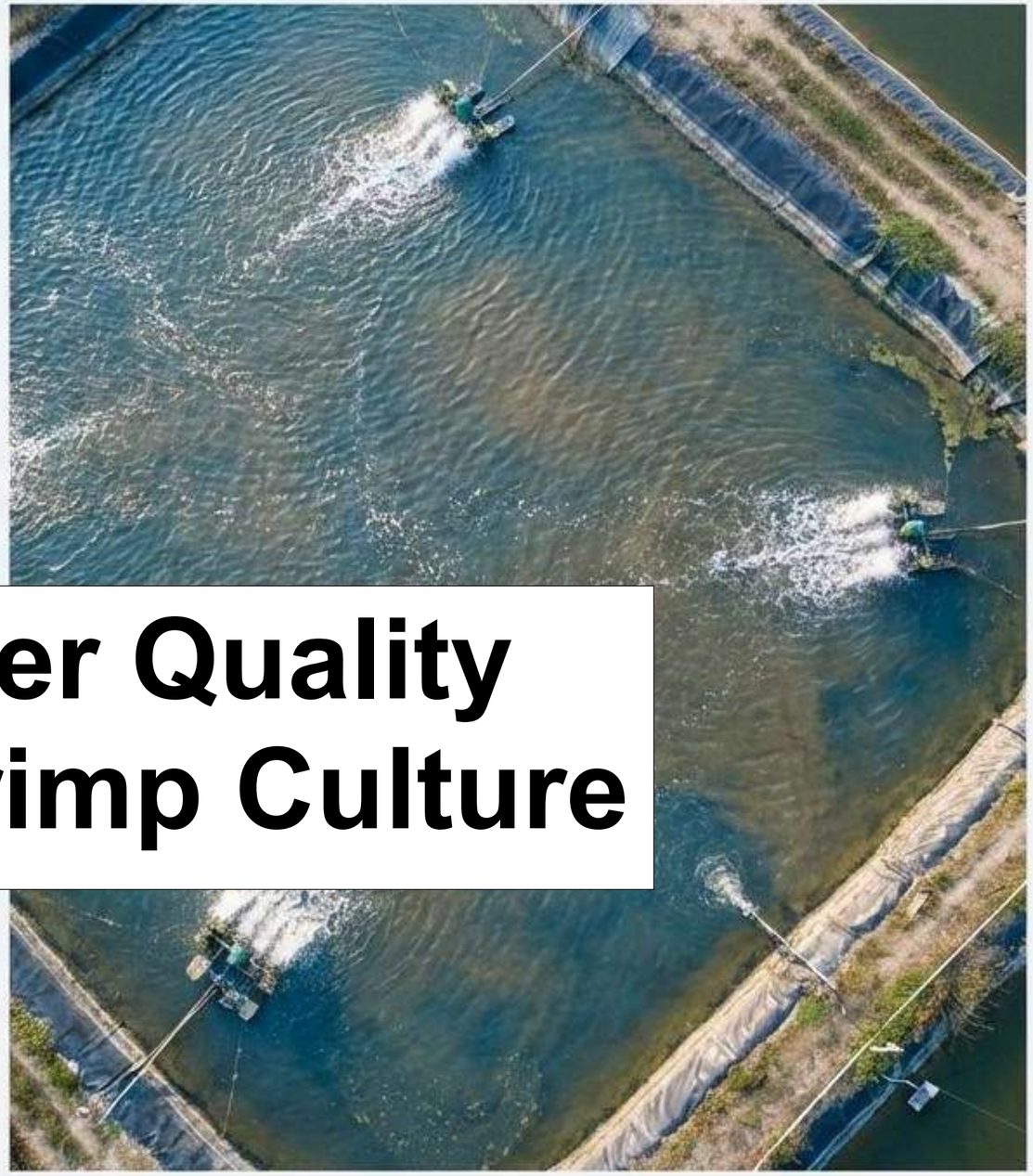
Analysis of Water Quality Parameters in Shrimp Culture

Bridging probability logic and water quality optimization through statistical inference.



DECEMBER 5, 2022 | STATISTICAL ANALYSIS & STRATEGIC RECOMMENDATIONS

The Reality of Biological Data: High Noise, Low Signal

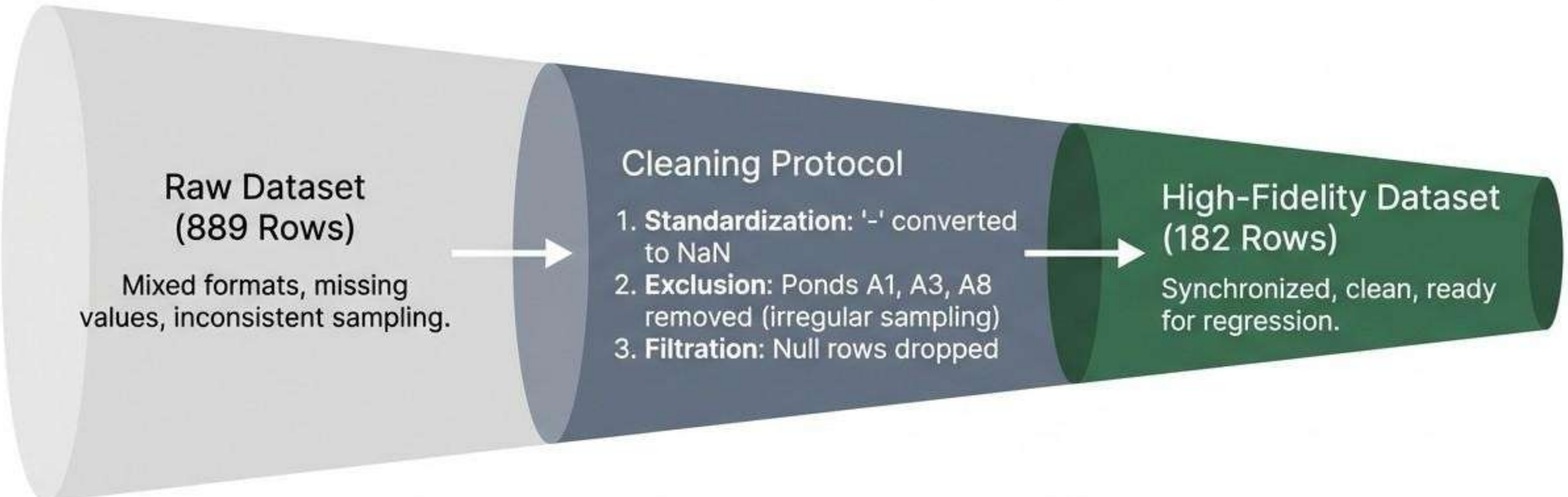
Date	Pond ID	TAN	Phosphate	Temp
2023-10-01	A1	NaN	0.5	-
2023-10-02	A2	1.2	NaN	28.5
2023-10-03	A3	-	-	NaN
2023-10-04	A4	NaN	0.5	29.5
2023-10-05	A5	[illegible]	[illegible]	NaN
2023-10-07	A6	NaN	0.6	28.5
2023-10-08	A7	NaN	NaN	28.5
2023-10-09	A8	NaN	0.4	29.5
2023-10-10	A9	NaN	0.5	28.5
2023-10-11	A0	NaN	NaN	NaN

72.1% Missing Values

- **Total Records:** 889 rows across 10 Ponds (A1–A9, AA10).
- **Inconsistency:** Ponds A1, A3, and A8 contained only 77 days of data vs 94 days for others.
- **Data Integrity:** Mixed use of 'NaN' and '-' required extensive pre-processing.

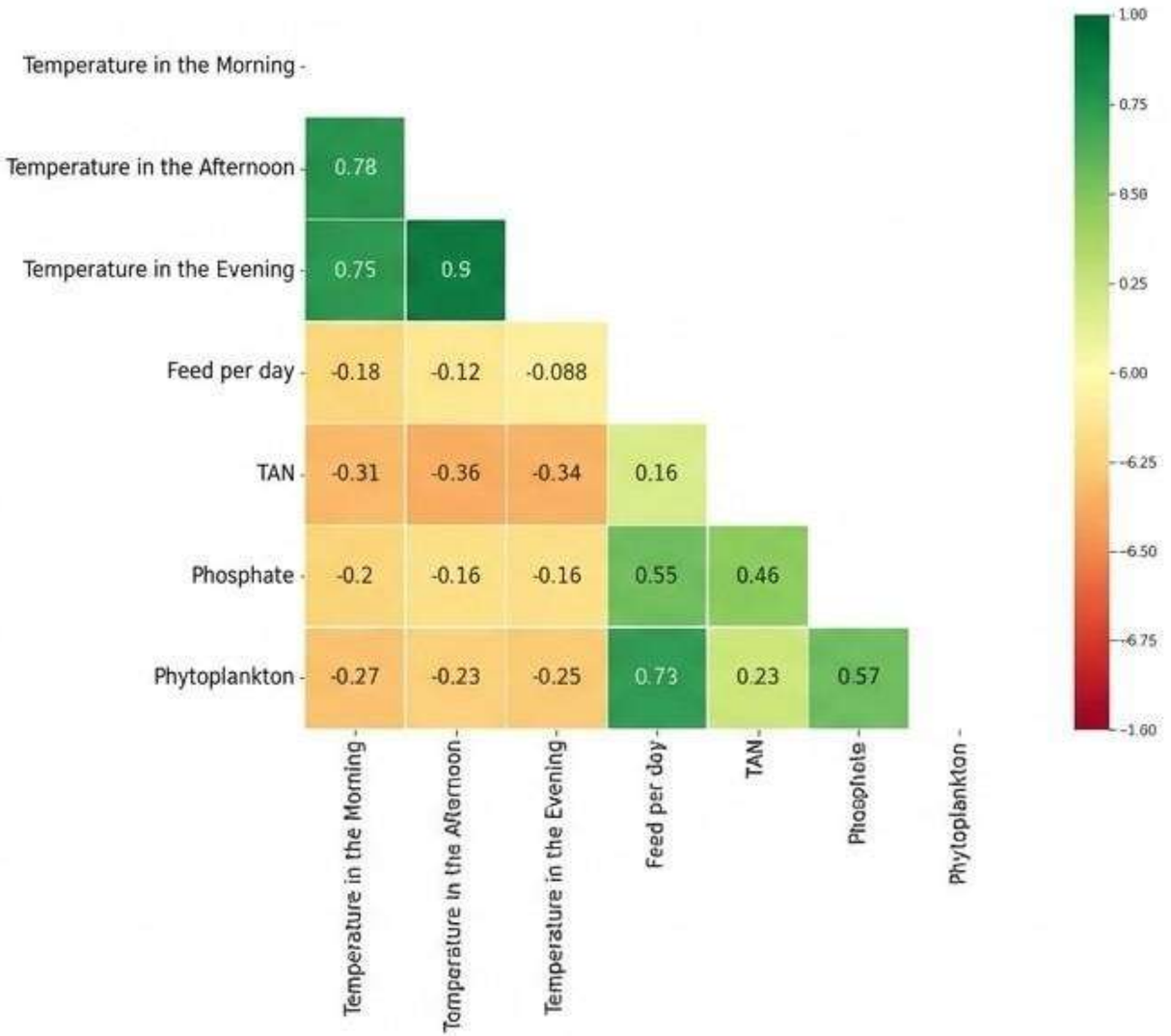
Methodology: Analysis based on 889 raw records across 10 ponds, refined to a high-fidelity dataset of 182 synchronized observations.

From Raw Data to Analytical Baseline



"We removed inconsistent grouped data records to ensure all analyzed ponds shared the same dimension and treatment conditions."

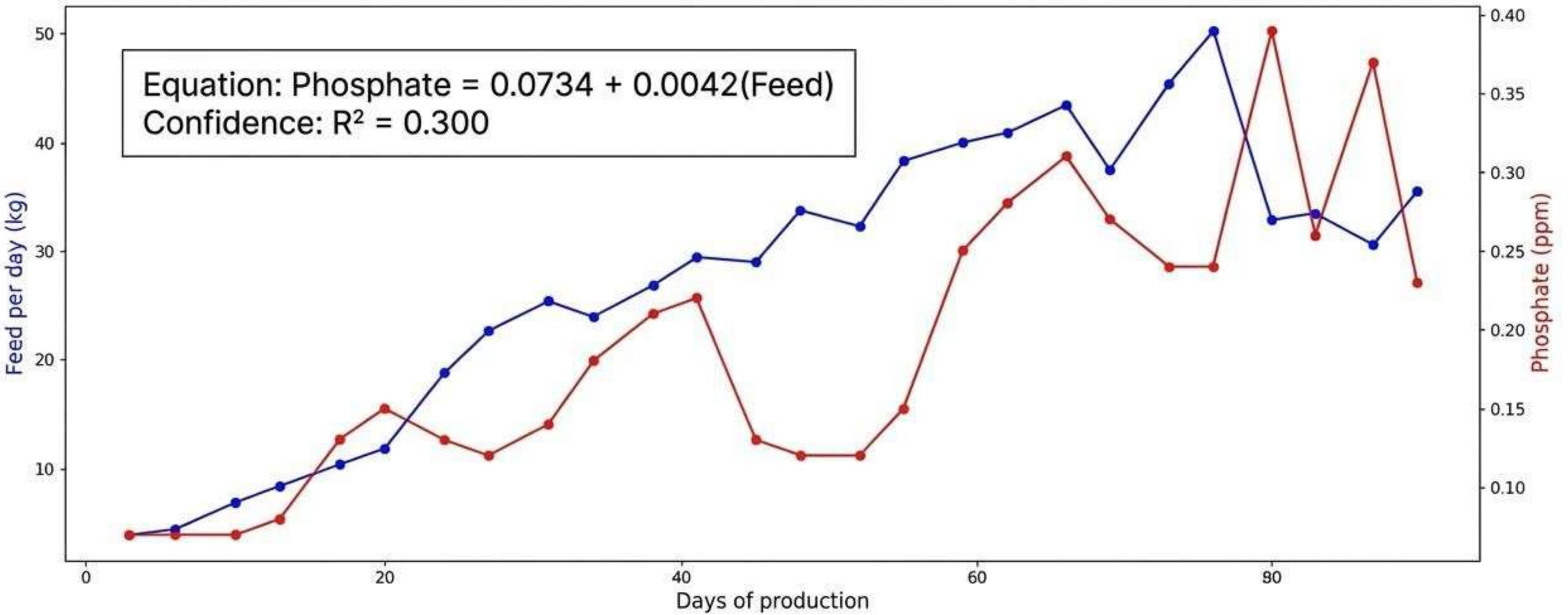
Mapping the Ecosystem: The Correlation Matrix



Key Relationships

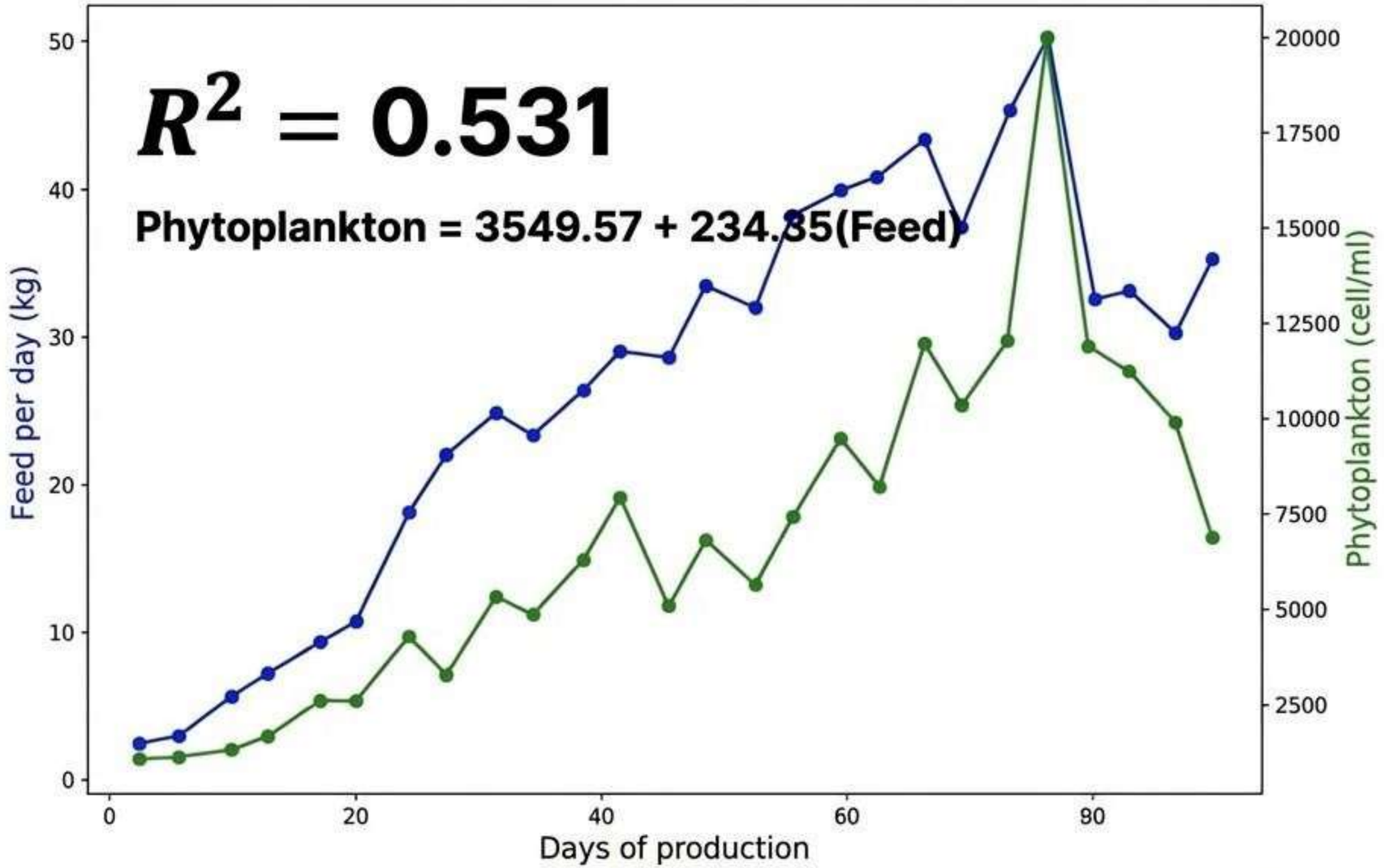
- Feed & Phytoplankton (0.73): Strong positive correlation. Feed acts as fertilizer.
- Feed & Phosphate (0.55): Moderate positive correlation.
- Temperature & TAN (-0.31): Negative correlation. Heat reduces toxicity.
- Feed & TAN (0.16): Weak direct correlation, suggesting biological mediation is required.

Feed Input Drives Linear Phosphate Accumulation



Insight: Feed input explains 30% of the variance in Phosphate. Every kg of feed adds a predictable, linear chemical load to the water.

Feed is the Primary Fuel for Phytoplankton Blooms

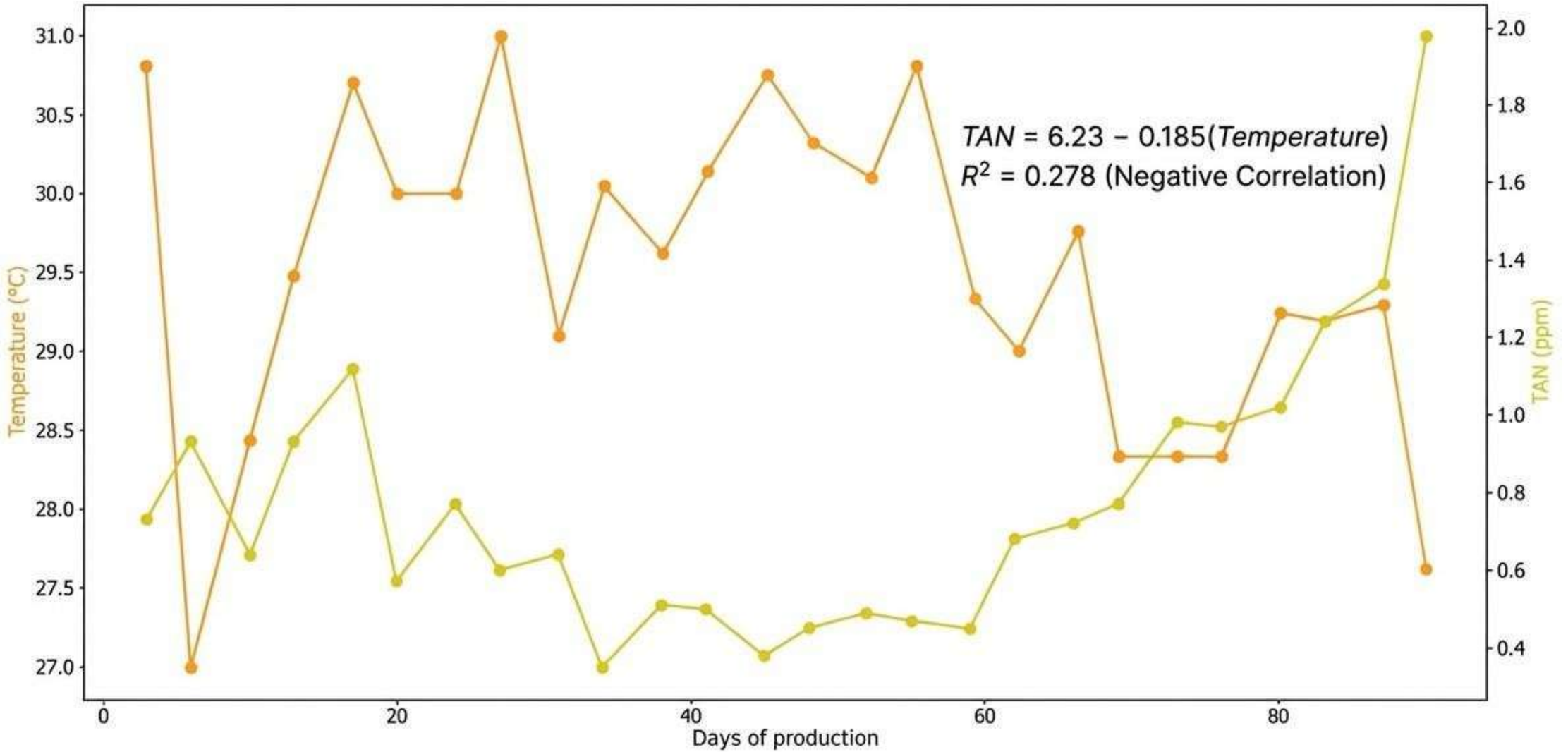


Key Insight

Strongest Correlation observed. Over 53% of variation in plankton density is directly attributable to feed quantity.

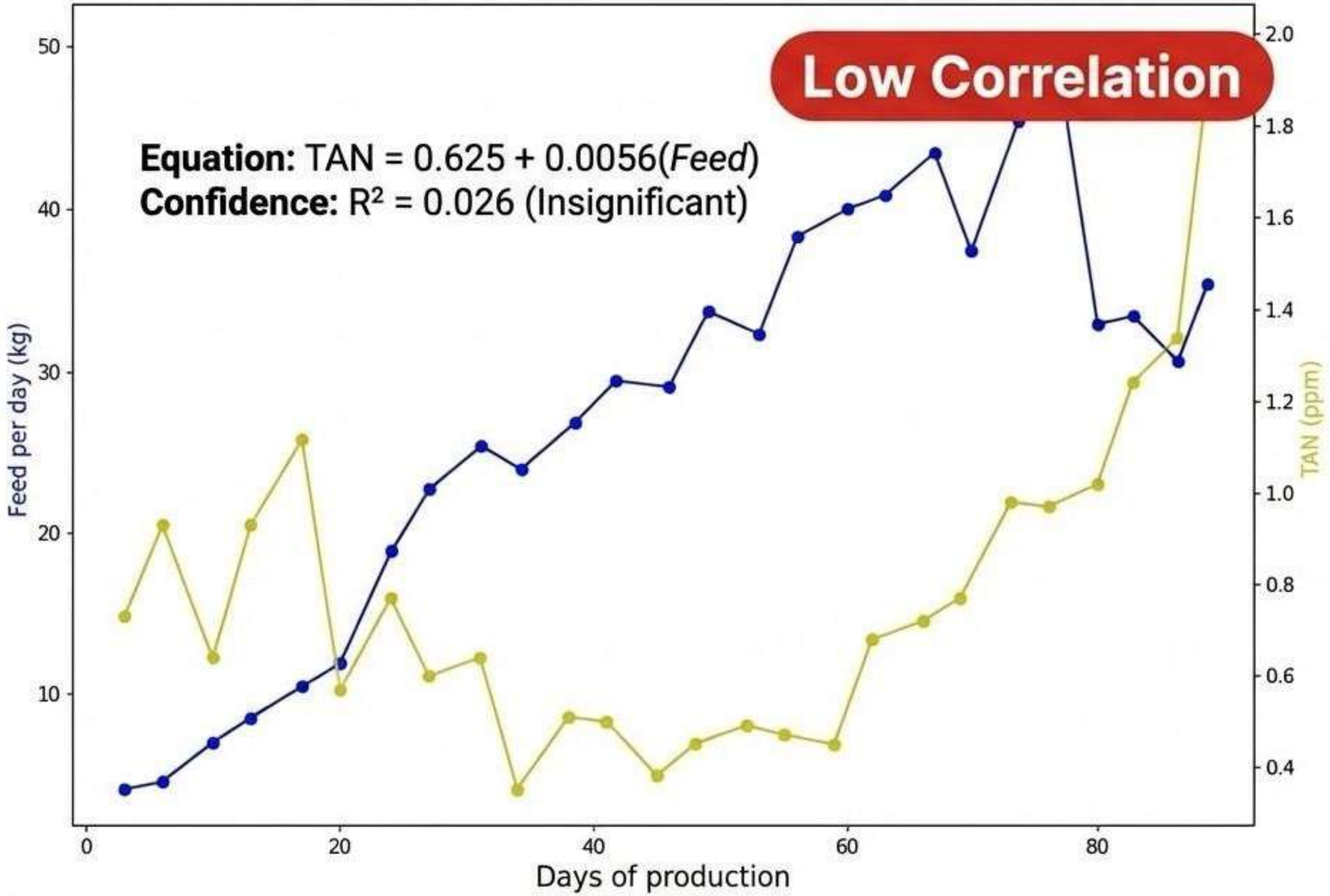
Managing feed is effectively managing the bloom.

The Thermal Advantage: Heat Accelerates Ammonia Degradation



Biological Driver: As water temperature approaches the optimal 29–30°C range, bacterial metabolism increases, speeding up the degradation of toxic ammonia.

The Missing Link: Why Feed Doesn't Directly Predict Ammonia

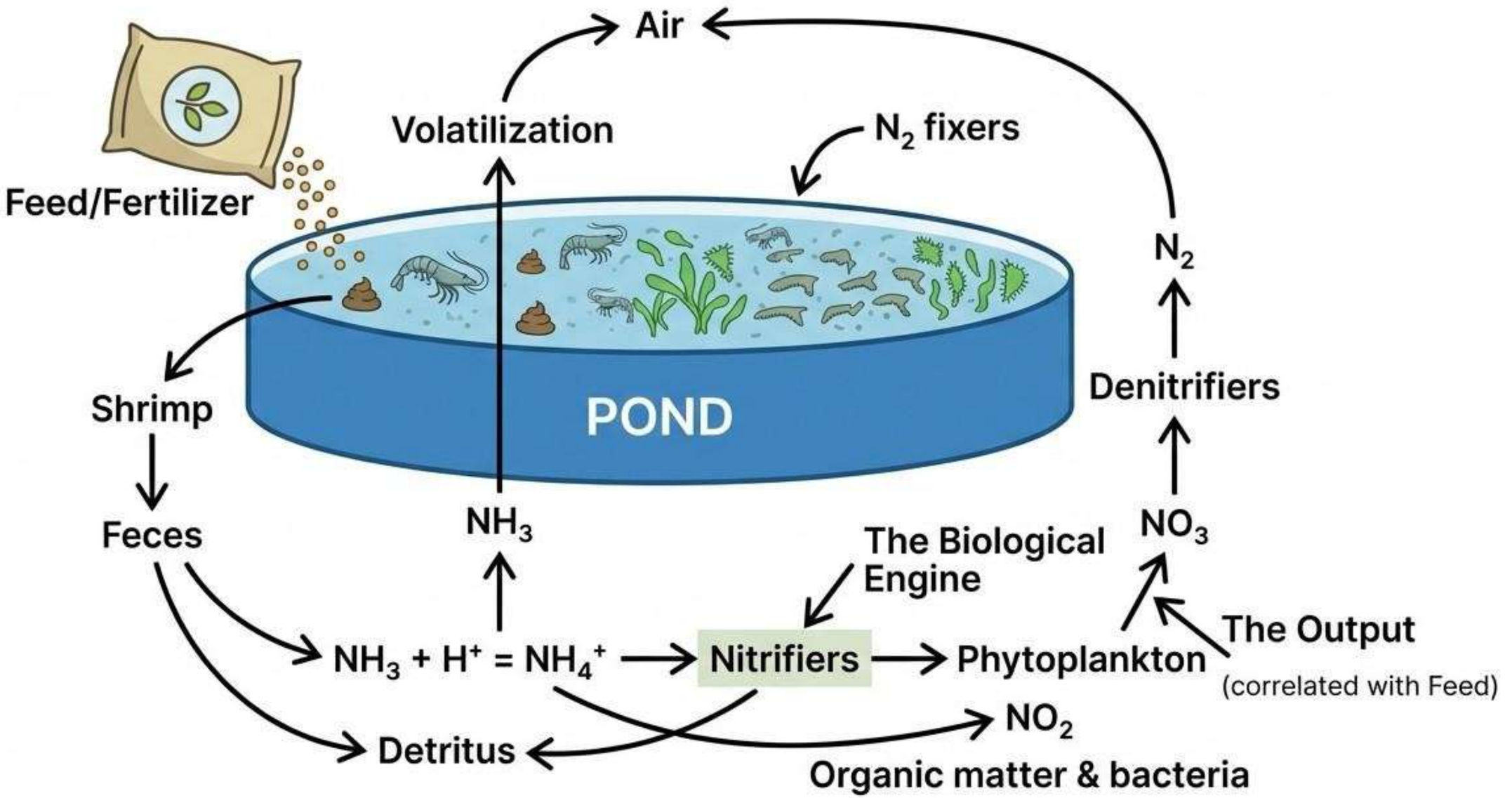


The Puzzle:

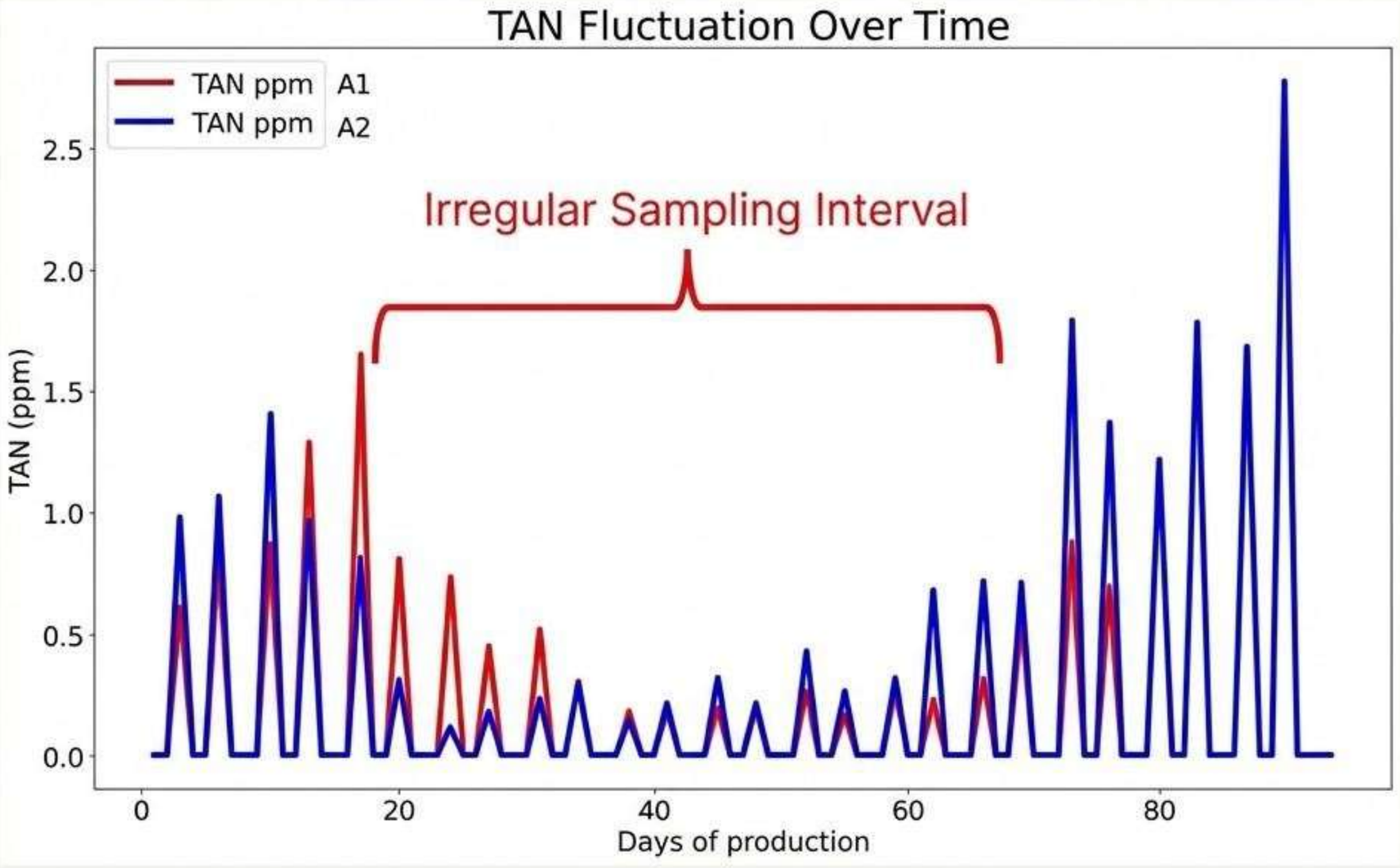
Feed contains nitrogen, yet there is no direct linear path to Ammonia.

This indicates a complex biological mediation process—The Nitrogen Cycle—is buffering the relationship.

The Mechanism: The Nitrogen Cycle in Intensive Ponds



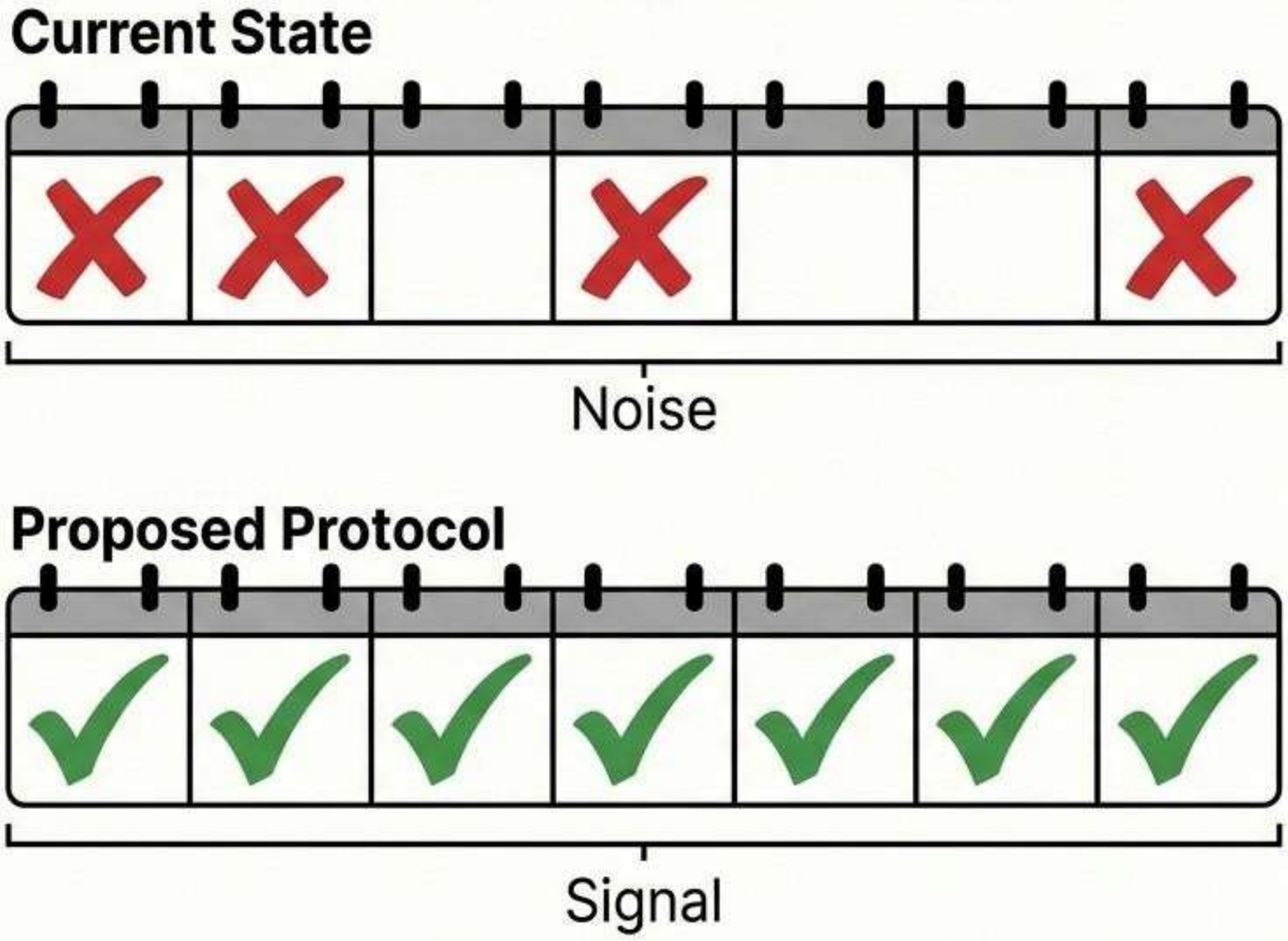
Critique of Current Data Architecture: The Sampling Gap



The Problem: Inconsistent measurement windows (2 days, 2 days, 3 days...) create artificial gaps.

Consequence: This 'Sawtooth' pattern prevents advanced Time Series Analysis and predictive toxicity modeling.

Strategic Recommendation 1: Standardized Data Protocol



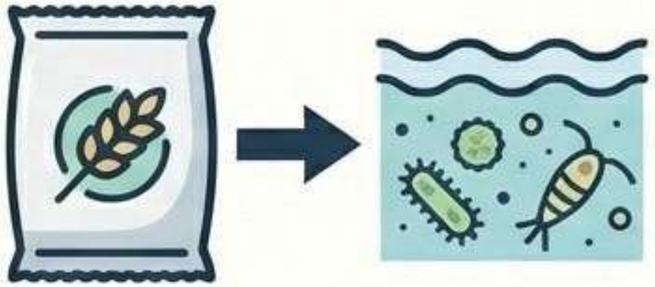
Objective: Enable Predictive ML & Forecasting.

Action: Implement strict 3-day or Weekly sampling windows across all ponds.

Handling Gaps: Use K-Nearest Neighbor (KNN) imputation only on consistent time-series backbones.

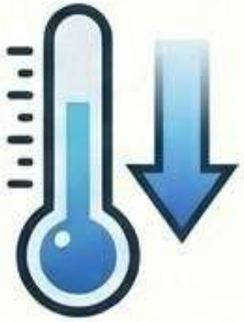
Strategic Recommendation 2: Nitrogen-Conscious Feeding

Precision Feeding



Shift from reactive to precision feeding. Feed is the primary driver of Phosphate and Plankton density. Overfeeding directly degrades water quality.

Temperature Awareness



Monitor for thermal drops. When temperature falls, bacterial metabolism slows (R^2=0.278), reducing the pond's ability to process ammonia. Reduce feed inputs during cold spells to prevent toxicity spikes.

Conclusion: From Reactive Observation to Causal Control

- **TAN Dynamics:** Highly volatile (0.35 - 1.98 ppm). Requires thermal regulation.
- **Phosphate Dynamics:** Stable accumulation. Linearly driven by feed input.
- **The Mechanism:** The Nitrogen Cycle mediates the conversion of feed to toxicity.

By standardizing data collection, DELOS can move from describing what happened to predicting what will happen, unlocking the potential for Machine Learning in aquaculture.